

AP EAMCET
Engineering Entrance Exam
24th April 2018- Shift 2

Physics

1. If $10 \text{ gcms}^{-1} = x \text{ Ns}$. then the number x is
- _____
- A. 1×10^{-5}
B. 1×10^{-4}
C. 1×10^{-6}
D. 1×10^{-3}
2. Assertion (A) : The displacement of a vertically projected body during last second of its upward motion is $\frac{g}{2}$
- Reason (R) : For a vertically projected body the acceleration decreases gradually and g becomes $\frac{g}{2}$ during the last second of upward motion.
- A. Both (A) and (R) are true and (R) is the correct explanation of (A)
B. Both (A) and (R) are true but (R) is not the correct explanation of (A)
C. (A) is true but (R) is false (A)
D. is false but (R) is true
3. A body is projected with a velocity 10 ms^{-1} at an angle of projection 60° from a point. If it reaches a point A such that

OA is $3\sqrt{3}\text{m}$ and OA makes 45° with the horizontal. then the time taken by the body to reach the point A is

- A. 0.21 s
- B. 0.42 s
- C. 0.52 s
- D. 0.64 s

4. A large number of bullets are fired in all directions with the same speed 'u'. The maximum area of the ground on which these bullets spread is (g - acceleration due to gravity)

- A. $\frac{\pi u^2}{g}$
- B. $\frac{\pi u^4}{g^2}$
- C. $\frac{\pi u^2}{g^4}$
- D. $\frac{\pi u}{g}$

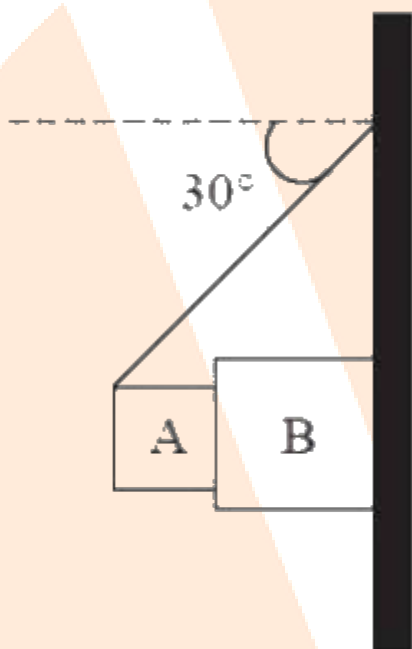
5. Two blocks of masses m and 2m kept on a frictionless horizontal surface are connected by a massless string. Two horizontal forces $F_1 = (4.2t)\text{ N}$ and $F_2 = (7.5t)\text{ N}$, where 't' is time in seconds, are acting on the system as shown in the figure. The time at which the tension in the string between the two blocks becomes 10.6 N is s.



- A. 4
- B. 3
- C. 2

D.6

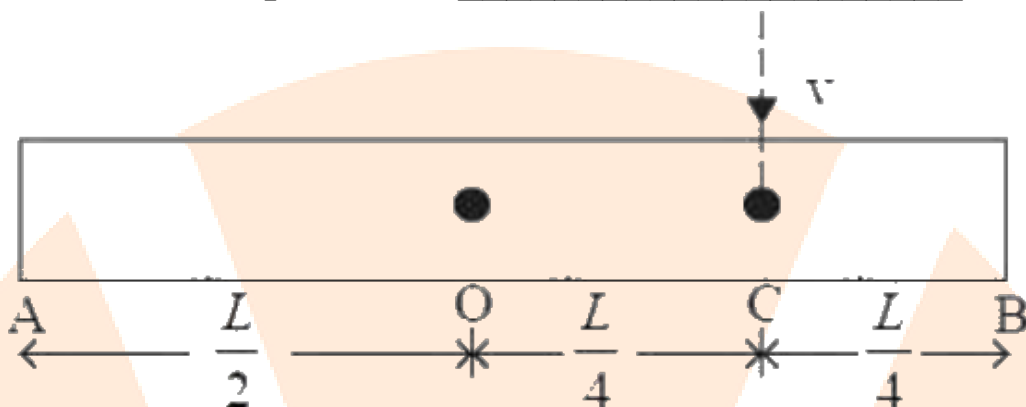
6. Two blocks A and B of masses 5 kg and 10 kg respectively are placed as shown in the figure. The coefficient of friction between all the surfaces is 0.2. Then the acceleration of the block B is ($g = 10 \text{ ms}^{-2}$)



- A. 2.6 ms^{-2}
 B. 46 ms^{-2}
 C. 26 ms^{-2}
 D. 4.7 ms^{-2}
7. A ball moving horizontally with a velocity 2 ms^{-1} strikes the bob of a seconds pendulum at rest. If the mass of the bob is equal to the mass of the ball and the collision is perfectly elastic. after collision the bob of the pendulum will raise to a height of ($g = 10 \text{ ms}^{-2}$)
- A. 80 cm
 B. 60 cm

- C. 40 cm
D. 20 cm
8. A body of mass 2 kg moving in X-Y plane has a potential energy given by $U = (6x + 8y)$ J. The body is at rest at the point (3, 2) m. The work to be done by the body to reach another position after 2 s is
- A. 100 J
B. 500 J
C. 750 J
D. 900 J
9. A ball of mass 2 kg collides with another ball of mass M at rest. If the collision is elastic and after the collision, the first ball moves with $\frac{1}{3}$ of its initial velocity in the same direction, then the mass of second ball is
- A. 2.5 kg
B. 3.5 kg
C. 3.0 kg
D. 1.0 kg
10. A uniform rod AB of length 'L' and mass 'M' is pivoted at the centre 'O' in such a way that it can rotate in the vertical plane as shown in the figure. An insect of the same mass falls vertically with speed 'v' on the horizontal rod at a point 'C' which is the midpoint of O and B. Immediately after falling, the insect moves towards the end B such that the rod rotates with constant angular velocity ' ω '. If the

insect reaches the end B when the rod has turned through 90° , then the speed V is _____



- A. $\frac{12}{7}gL$
- B. $\frac{7}{12}\sqrt{2gL}$
- C. $\frac{7}{12}\sqrt{gL}$
- D. $\frac{12}{7}\sqrt{gL}$

11. A particle starts executing simple harmonic motion from one extreme position. If a , b and c are the displacements of the particle from the mean position at the ends of three successive seconds. the frequency of simple harmonic motion is _____

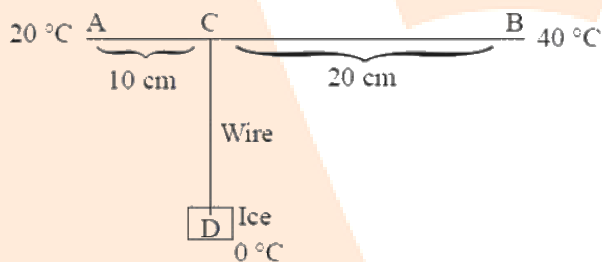
- A. $\frac{1}{\pi} \cos^{-1} \left[\frac{a+b}{c} \right]$
- B. $\frac{1}{2\pi} \cos^{-1} \left[\frac{b+c}{2a} \right]$
- C. $\frac{1}{2\pi} \cos^{-1} \left[\frac{a+c}{b} \right]$
- D. $\frac{1}{2\pi} \cos^{-1} \left[\frac{a+b}{2c} \right]$

12. A satellite is to be placed in equatorial geostationary orbit around the earth for communication purpose. The height of such satellite is _____ ($M_E = 6 \times 10^{24} \text{ kg}$, $R_E = 6400 \text{ km}$)
- A. $3.57 \times 10^8 \text{ m}$
 - B. $3.57 \times 10^7 \text{ m}$
 - C. $3.57 \times 10^5 \text{ m}$
 - D. $3.57 \times 10^6 \text{ m}$
13. For a material, the ratio of Young's modulus to rigidity modulus is 2.8. If some force is applied on a wire made of this material, its cross-sectional area decreases by 2%. The percentage change in its length is
- A. 0.4
 - B. 1.6
 - C. 2.5
 - D. 4
14. A liquid is filled in a tank up to a height of 5 in and the bottom of the tank is at 5 m height from the ground. A hole is punched to the tank so that the liquid coming out touches the ground at maximum horizontal distance. Then the maximum horizontal distance is
- A. 10 m
 - B. 2.5 m
 - C. 3.5 m
 - D. 15 m
15. A solid sphere of lithium is rotating with angular frequency ' ω ' about an axis passing through its diameter. If its

temperature is raised by 50°C then its new angular frequency is ($\alpha_{\text{lithium}} = 60 \times 10^{-5} \text{ }^{\circ}\text{C}^{-1}$)

- A. 0.99ω
- B. 0.73ω
- C. 0.83ω
- D. 0.94ω

16. In the figure shown. AB is a rod of length 30 cm. area of cross-section 1 cm^2 and thermal conductivity 336 SI units. The ends A and B are at constant temperatures 20°C and 40°C respectively. A point C of the rod is connected to ice at 0°C in a thermally insulated box D through a highly conducting wire of negligible heat capacity. The rate at which ice melts in the box is ($L_{\text{ice}} = 80 \text{ cal g}^{-1}$)



- A. 84 mgs^{-1}
 - B. 84 gs^{-1}
 - C. 20 mgs^{-1}
 - D. 40 mgs^{-1}
17. Hydrogen gas is undergoing a process given by $PV^2 = \text{constant}$. The ratio of work done by the gas to the change in its internal energy is _____
- A. $\frac{2}{3}$
 - B. 0.4

C. 1.5

D. -0.4

18. 2.8 g of nitrogen gas is in a vessel at a temperature 127°C . The amount of heat energy required to increase the rms speed of the nitrogen molecules by 41.4% is _____ ($R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}$)

A. 376 J

B. 415 J

C. 1662 J

D. 831 J

19. A gaseous mixture has 2 moles of oxygen and 4 moles of Argon at a temperature T . Neglecting all vibrational modes of the molecules, the total internal energy of the system is _____ (R -Universal gas constant)

A. $4RT$

B. $15RT$

C. $9RT$

D. $11RT$

20. A motor cycle starts from rest from a stationary source of sound and moves away from the source with a uniform acceleration 2 ms^{-2} . Distance travelled by the motor cycle when the person on it hears the sound of frequency which is 94% of the true frequency, is nearly (speed of sound in air = 330 ms^{-1})

A. 1.49 m

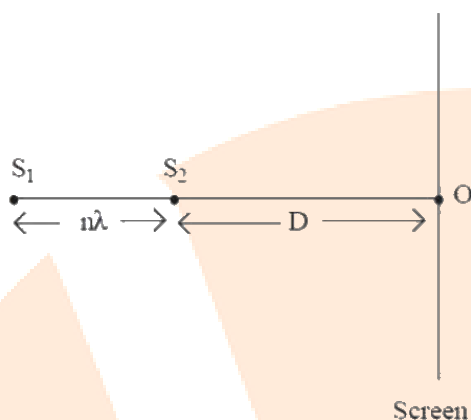
B. 98 m

C. 147 m

D. 196 m

21. A steel wire of length 1 mass 0.1 kg and uniform area of cross section 10^{-6}m^2 is rigidly fixed at both the ends without any tension. Its temperature is lowered by 20°C and transverse waves are set up by plucking the wire at the middle. The frequency of the fundamental mode is ($Y = 200\text{ GPa}$, $\alpha = 1.21 \times 10^{-5}^\circ\text{C}^{-1}$)
- A. 11 Hz
 - B. 42 Hz
 - C. 11 Hz
 - D. 22 Hz
22. For a prism. the angle of the prism is 60° and the refractive index is $\sqrt{\frac{7}{3}}$. The minimum possible angle of incidence so that the light ray is refracted from the second surface is
- A. 15°
 - B. 25°
 - C. 30°
 - D. 35°
23. Two coherent sources S_1 , S_2 and a screen are arranged as shown in the figure. If the distance between the two coherent sources is $n\lambda$ and the distance of the screen from the nearer coherent source is D . then the distance of the first bright fringe on the screen from the point 'O' is (λ X is

the wavelength of the light emitted by the coherent sources



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A. $\sqrt{\frac{D(D+n\lambda)}{n}}$

B. $\sqrt{\frac{2D(D+n\lambda)}{n}}$

C. $\sqrt{\frac{3D(D+n\lambda)}{n}}$

D. $\sqrt{\frac{D(D+n\lambda)}{2n}}$

24. Two charged particles of masses in the ratio 1 : 3 have charges in reciprocal ratio as their masses. They are placed in a uniform electric field and allowed to move. The ratio of their kinetic energies is

A. 9 : 1

B. 4 : 3

C. 3 : 1

D. 27 : 1

25. Two point charges $-5\mu\text{C}$ and $+5\mu\text{C}$ are placed at A and B respectively on a straight line separated by 5 cm. At a point C if the resultant field is parallel to the line joining the points A and B then

A. $AC = BC$

B. $2AC = BC$

C. $AC = 2BC$

D. $AC = BC$

26. Match the following List-I with the List-II

List - I List – II

A) Gauss's law

I) Surface charge density

B) Faraday's law
conservation

II) Electric charge and energy

C) Ampere's law

III) Change in magnetic flux

D) Kirchhoff's law

IV) Change in electric flux

V) Total electric flux

A B C D

II III I IV

A B C D

IV III II I

A B C D

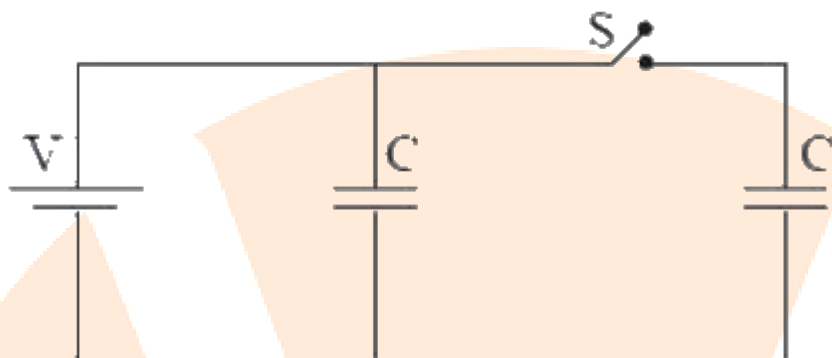
V III IV II

A B C D

V I IV III

27. Two identical condensers are joined as shown in the figure. When the switch S is closed, the total energy of the system is U_1 . If the switch is opened and both the condensers are filled with a dielectric of dielectric constant 3, then the energy of the system becomes U_2 .

The value of $\frac{U_1}{U_2}$ is



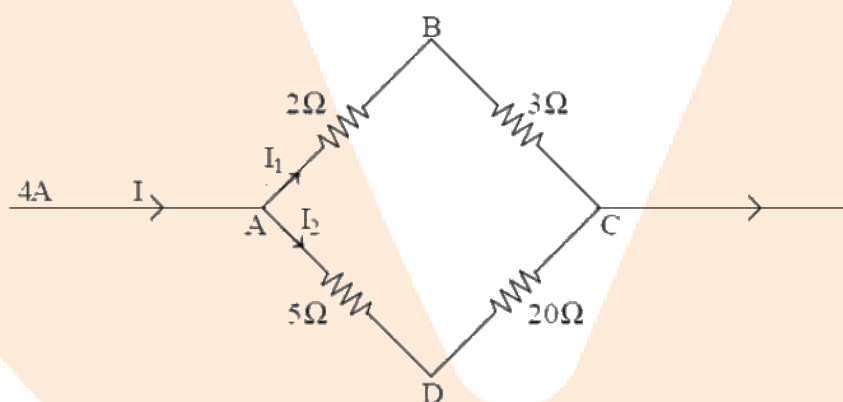
A. 3 : 1

B. 5 : 1

C. 3 : 5

D. 5 : 3

28. In the given network. the potential difference between the points B and D is



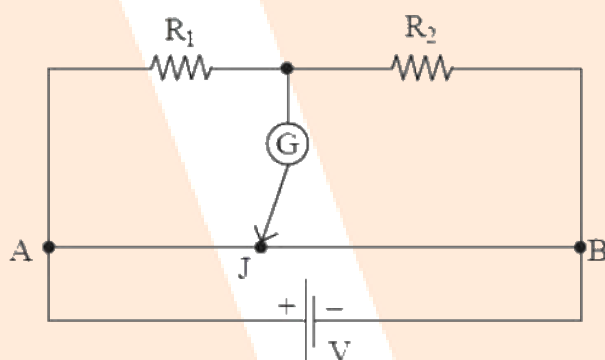
A. $-\frac{10}{3}V$

B. $-\frac{20}{3}V$

C. $\frac{4}{3}V$

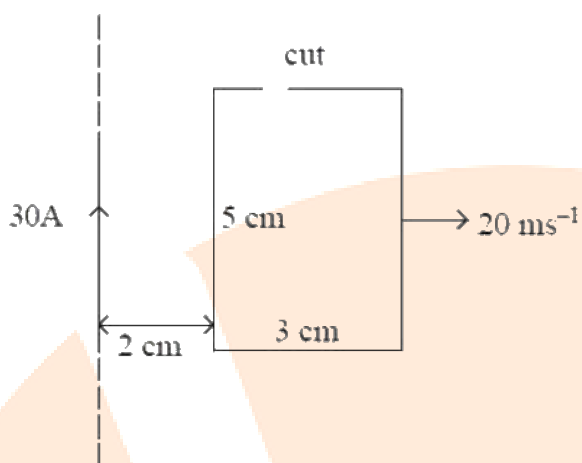
D. $\frac{2}{3}V$

29. A uniform wire of length 200 cm is connected to a battery, two resistors and a galvanometer as shown in the figure. The galvanometer shows null deflection when the jockey J is at 80 cm from point A. If the resistor R^2 is shunted with $30\ \Omega$ resistance, the galvanometer shows null deflection when the jockey J is at 100 cm from point B. Then the values of R^1 and R^2 respectively are



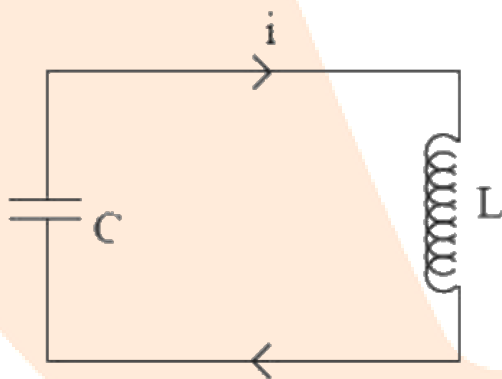
- A. $20\ \Omega$, $30\ \Omega$
B. $30\ \Omega$, $30\ \Omega$
C. $15\ \Omega$, $10\ \Omega$
D. $10\ \Omega$, $15\ \Omega$
30. The sensitivity of a galvanometer is $60\ \text{div}/\text{A}$. When a shunt is used, its sensitivity becomes $10\ \text{div}/\text{A}$. If the resistance of the galvanometer is $20\ \Omega$, then the value of shunt used is
- A. $4\ \Omega$
B. $5\ \Omega$
C. $20\ \Omega$
D. $2\ \Omega$

31. A thin wire of length L made of an insulating material is bent to form a circular loop and a positive charge 'q' is given so that it is distributed uniformly around the circumference of the loop. The loop is then rotated with an angular speed ' ω ' about an axis passing through its centre. If a uniform magnetic field B directed parallel to the plane of the loop is applied then the magnitude of the magnetic torque on the loop is
- $\frac{q\omega L^2 B}{8\pi^2}$
 - $\frac{q\omega L^2 B}{4\pi^2}$
 - $\frac{q\omega L^2 B}{2\pi^2}$
 - $\frac{q\omega L^2 B}{\pi^2}$
32. A short magnet oscillates with a frequency 10 Hz where horizontal component of Earth's magnetic field is $12 \mu\text{T}$. A downward current of 15 A is established in a vertical wire placed 20 cm west of the magnet. The new frequency of the magnet is _____
- 4 Hz
 - 2.5 Hz
 - 9 Hz
 - 5 Hz
33. A rectangular wire loop of sides 5 cm and 3 cm with a small cut is moving away from an infinitely long straight wire carrying a current of 30 A with a velocity of 20 ms^{-1} as shown in the figure. The emf induced across the cut is



- A. $50\text{ }\mu\text{V}$
- B. $75\text{ }\mu\text{V}$
- C. $180\text{ }\mu\text{V}$
- D. $150\text{ }\mu\text{V}$

34. In the given L-C circuit. inductance is $80\text{ }\mu\text{H}$ and capacitance is $20\text{ }\mu\text{F}$ The maximum potential difference across the capacitor is 80 V . The maximum current is



- A. 200 A
- B. 40 A
- C. 800 A
- D. 100 A

35. An electromagnetic wave of wavelength $2 \times 10^{-10} \text{ m}$ passes from vacuum into a dielectric medium of relative permittivity 4. Then its wavelength will be
- $0.5 \times 10^{-10} \text{ m}$
 - $0.7 \times 10^{-10} \text{ m}$
 - $1.8 \times 10^{-10} \text{ m}$
 - $2.8 \times 10^{-10} \text{ m}$
36. When photons of energy 5 eV incident on a photo metal of work function 4.36 eV. photo electrons are emitted. The maximum impulse of the metal surface is _____
- $2.41 \times 10^{-25} \text{ kgms}^{-1}$
 - $2.31 \times 10^{-25} \text{ kgms}^{-1}$
 - $4.31 \times 10^{-25} \text{ kgms}^{-1}$
 - $1.31 \times 10^{-25} \text{ kgms}^{-1}$
37. A moving Hydrogen atom makes a head on collision with a stationary Hydrogen atom. Before collision both atoms are in ground state and after collision they move together. The minimum kinetic energy of the moving Hydrogen atom, such that one of the atoms reaches the excitation state is _____
- 13.6 eV
 - 30.6 eV
 - 20.4 eV
 - 10.2 eV
38. Activity of a radioactive sample is R_1 at a time t_1 and R_2 at a time t_2 . Its half life period is T . The number of atoms that have disintegrated in the time interval $(t_2 - t_1)$ is equal to

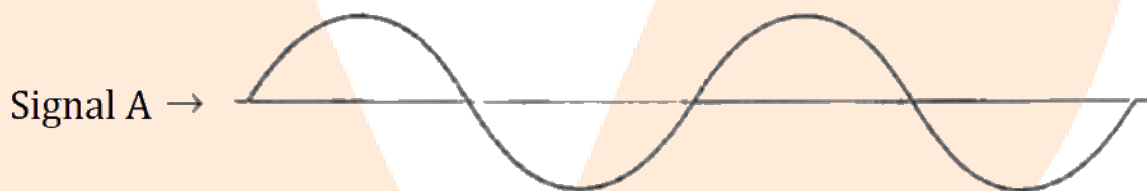
$\frac{n(R_1 - R_2)T}{\ln 4}$. Then 'n' is equal to

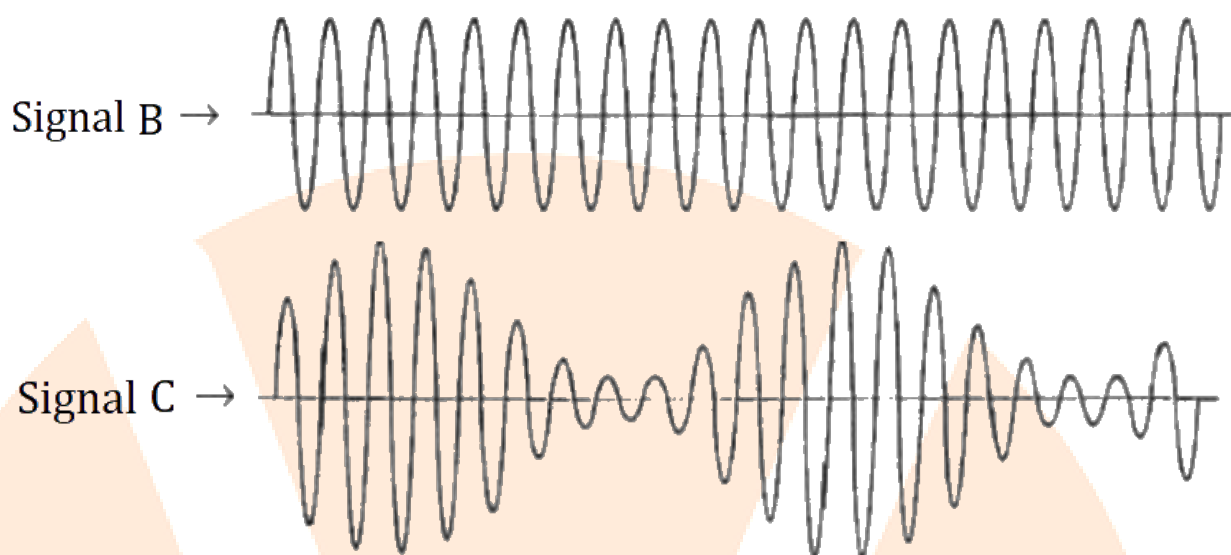
- A. 2
- B. 1
- C. 4
- D. $\frac{1}{2}$

39. In a common emitter amplifier, the output resistance is 5000Ω and the input resistance is 2000Ω . If the peak value of input signal voltage is 100 mV and $\beta = 50$, then the peak value of output voltage is

- A. $5 \times 10^{-6} \text{ V}$
- B. $12.5 \times 10^{-4} \text{ V}$
- C. 12.5 V
- D. 1.25 V

40. Three signals related to amplitude modulation are shown in the figure. The correct assignment of the signals is





- A. Signal A : information message. Signal B : Amplitude modulation
- B. Signal A : information message. Signal C : Amplitude modulation
- C. Signal B : Carrier wave. Signal C : Information message
- D. Signal B : Information message. Signal C : Amplitude modulation